

Data Mining Project

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Predicting and Understanding Obesity: A Data Mining Approach

Executive Summary

This project analyzed lifestyle and health factors influencing Body Mass Index (BMI) and obesity using a dataset of 2,111 individuals. Key findings reveal that eating between meals (EFBM) and history of being overweight (HoV) are the strongest predictors of obesity. Cluster analysis identified four distinct lifestyle groups, with only 17% exhibiting consistently healthy patterns. Recommendations focus on promoting physical activity, reducing fizzy drink consumption, and encouraging active transportation.

Business Understanding

Problem Statement

Obesity is a growing health crisis. This project aims to identify factors influencing BMI and obesity to support targeted interventions.

Project Objectives

1. Identify factors influencing BMI

2. Determine factors influencing obesity classification
3. Profile individuals across six obesity levels
4. Categorize individuals by lifestyle and health habits

CRISP-DM Framework

This analysis follows the CRISP-DM methodology:



Figure 1: CRISP-DM methodology

Data Understanding

Dataset Description

The dataset contains 2,111 records with 17 attributes:

Statistics					
Age	Integer	0	Min 14	Max 61	Average 24.316
EVM	Integer	0	Min 1	Max 3	Average 2.423
NDMM	Integer	0	Min 1	Max 4	Average 2.688
WATER	Integer	0	Min 1	Max 3	Average 2.015
ACTIVITY	Integer	0	Min 0	Max 3	Average 1.007
UoT	Integer	0	Min 0	Max 2	Average 0.665
Gender	Polynomial	0	Least Female (1043)	Most Male (1068)	Values Male (1068), Female (1043)
HoV	Polynomial	0	Least no (385)	Most yes (1726)	Values yes (1726), no (385)
FHCF	Polynomial	0	Least no (245)	Most yes (1866)	Values yes (1866), no (245)
EFBM	Polynomial	0	Least no (51)	Most Sometimes (1765)	Values Sometimes (1765), Frequently (242), ...[2 more]
SMOKE	Polynomial	0	Least yes (44)	Most no (2067)	Values no (2067), yes (44)
MONITOR	Polynomial	0	Least yes (96)	Most no (2015)	Values no (2015), yes (96)
FIZZY	Polynomial	0	Least Always (1)	Most Sometimes (1401)	Values Sometimes (1401), no (639), ...[2 more]
TRANSPORT	Polynomial	0	Least Bike (7)	Most Public_T [...] on (1580)	Values Public_Transportation (1580), Automobile (457), ...[3 more]
Height	Real	0	Min 1.450	Max 1.980	Average 1.702
Weight	Real	0	Min 39	Max 173	Average 86.586

Figure 2: Dataset Statistical Summary

Data Preparation

BMI Calculation

BMI was calculated using:

$$BMI = Weight / (Height \times Height)$$

BMI Categories (6 Classes)

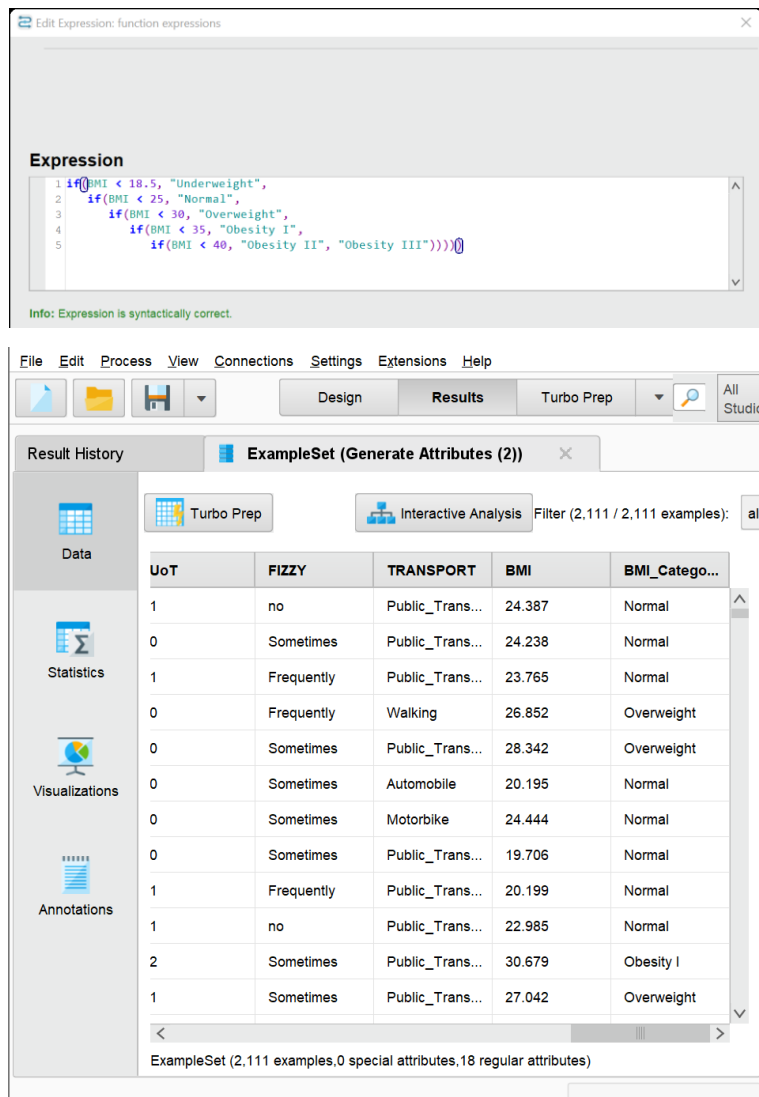


Figure 3: BMI Categories attribute generation

Ordinal Variable Conversion

EFBM and FIZZY were converted to numeric (1-4):

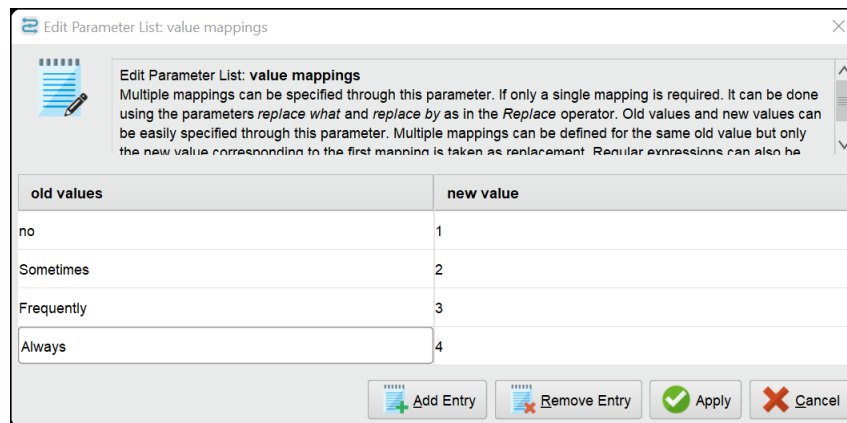


Figure 4: EFBM and FIZZY numeric conversion

Binary Variable Conversion

All Yes/No variables converted to 1/0 for regression modeling.

Adult Filtering

Only adults aged 20+ were included: **1,639 records** (removed 472 individuals under 20).

Result History

ExampleSet (Filter Examples)

Turbo Prep Interactive Analysis Filter (1,639 / 1,639 examples): all

Row No.	FIZZY	EFBM	Gender	Age
1	1	2	Female	21
2	2	2	Female	21
3	3	2	Male	23
4	3	2	Male	27
5	2	2	Male	22
6	2	2	Male	29
7	2	2	Female	23
8	2	2	Male	22
9	3	2	Male	24
10	1	2	Male	22
11	2	3	Male	26
12	2	3	Female	21

ExampleSet (1,639 examples, 0 special attributes, 18 regular attributes)

Figure 5: Age 20+ Filter applied

Modeling & Results

Question 1: Factors Influencing BMI (Linear Regression)

Business Understanding

Goal: Identify which lifestyle factors significantly affect continuous BMI values to understand what increases or decreases BMI.

Data Understanding

Target variable: BMI (continuous). Predictor variables: all lifestyle factors including EFBM, HoV, FHCF, Transport, FIZZY, and others.

Data Preparation

- BMI calculated from weight/height²
- Adults only (age ≥ 20 filter applied)
- Ordinal variables EFBM and FIZZY converted to numeric (1-4)
- Binary Yes/No variables converted to 1/0

Modeling

Linear regression with BMI as target variable and all lifestyle factors as predictors.

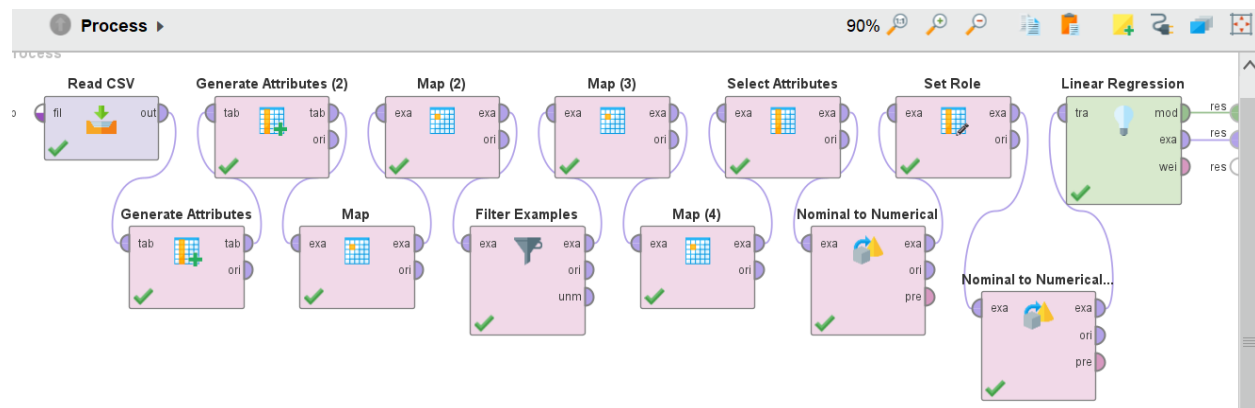


Figure 6: Process Diagram of Q1

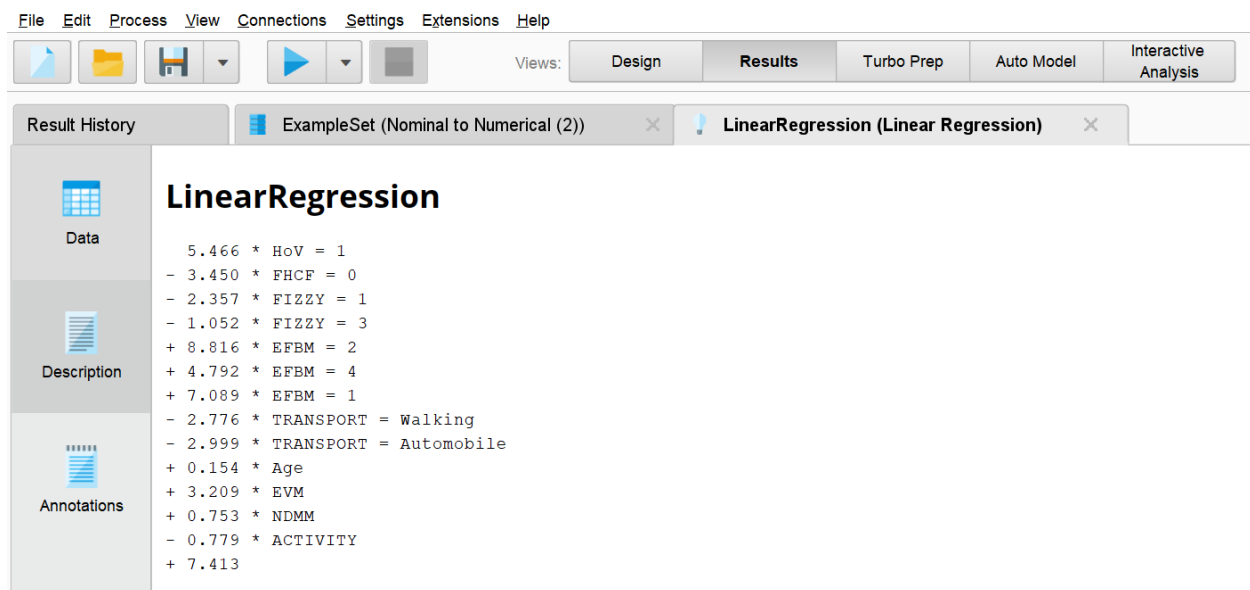


Figure 7: Linear regression Results

Evaluation

Factors INCREASING BMI

Factor	Coefficient	Interpretation
EFBM = Sometimes	+8.82	Strongest positive effect
EFBM = No	+7.09	Not eating between meals also increases BMI
HoV = Yes	+5.47	History of overweight increases BMI
EFBM = Always	+4.79	Always eating between meals increases BMI

Factors DECREASING BMI

Factor	Coefficient	Interpretation
FHCF = No	-3.45	No high-caloric food = lower BMI
Transport = Automobile	-3.00	Car use associated with lower BMI
Transport = Walking	-2.78	Walking decreases BMI
FIZZY = No	-1.05	No fizzy drinks = lower BMI

Conclusion for Q1

Eating habits (EFBM), weight history (HoV), and transportation choices are the strongest determinants of BMI.

Question 2: Factors Influencing Obesity (Decision Tree)

Business Understanding

Goal: Predict categorical obesity level (not continuous BMI) and rank predictor importance to identify key factors influencing obesity classification.

Data Understanding

Target variable: BMI Categories (6 classes: Underweight, Normal, Overweight, Obesity I, Obesity II, Obesity III). Predictor variables: same lifestyle factors as Q1.

Data Preparation

Same data preparation as Q1, with additional attention to class balance addressed through cross-validation.

Modeling

Decision tree classifier with 6-fold cross validation to predict BMI categories.

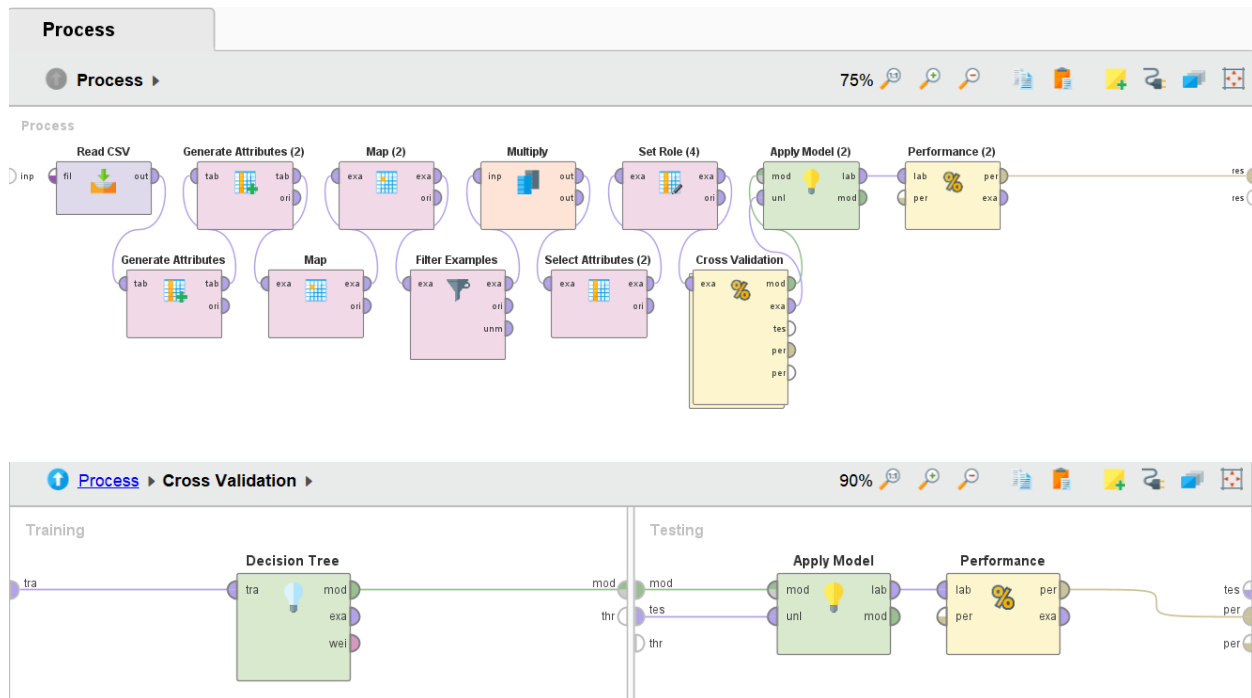


Figure 8: Process Diagram of Q2

Evaluation

- **Overall Accuracy: 47.90%**

accuracy: 47.90%

	true Normal	true Overweight	true Obesity I	true Underweight	true Obesity II	true Obesity III	class precision
pred. Normal	102	22	4	18	3	0	68.46%
pred. Overweight	9	93	3	11	9	0	74.40%
pred. Obesity I	3	5	6	1	0	0	40.00%
pred. Underweight	18	7	1	84	0	0	76.36%
pred. Obesity II	30	189	166	7	268	0	40.61%
pred. Obesity III	29	146	114	1	58	232	40.00%
class recall	53.40%	20.13%	2.04%	68.85%	79.29%	100.00%	

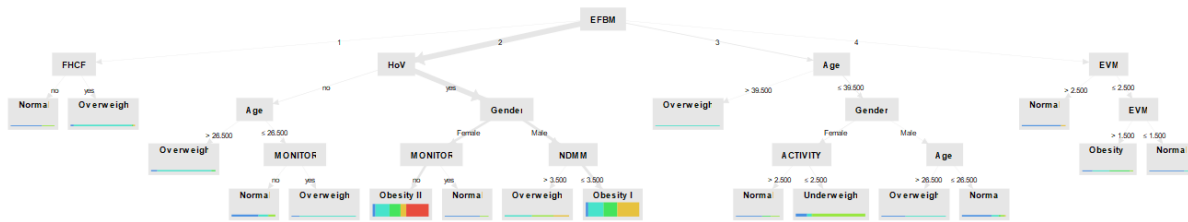


Figure 9: Confusion Matrix Analysis & Graph



Figure 10: Most Important Predictors (from Decision Tree)

**For EFBM = 2
(Sometimes):**

HoV = Yes, Male, NDMM $\leq$ 3.5 → **Obesity II** (268 cases)

HoV = Yes, Female, No calorie monitoring → **Obesity III** (232 cases)

**For EFBM = 3
(Frequently):**

Female, Low activity → **Underweight** (84 cases)

For EFBM = 1 (No):

FHCF = Yes → **Overweight** (36 cases)

Figure 11: Key Decision Tree Insights

Conclusion for Q2

Eating between meals is the single most important predictor of obesity status. History of overweight, gender, and calorie monitoring are also critical factors.

Question 3: Profiles of Six Obesity Levels

Business Understanding

Goal: Describe how lifestyle factors differ across each obesity class to understand the unique characteristics of individuals at each BMI level.

Data Understanding

Separate analysis by each of the six BMI categories: Underweight, Normal, Overweight, Obesity I, Obesity II, Obesity III.

Data Preparation

Filtered data by each BMI category and computed descriptive statistics for each group.

Modeling

Filtered data by each BMI category and computed descriptive statistics.

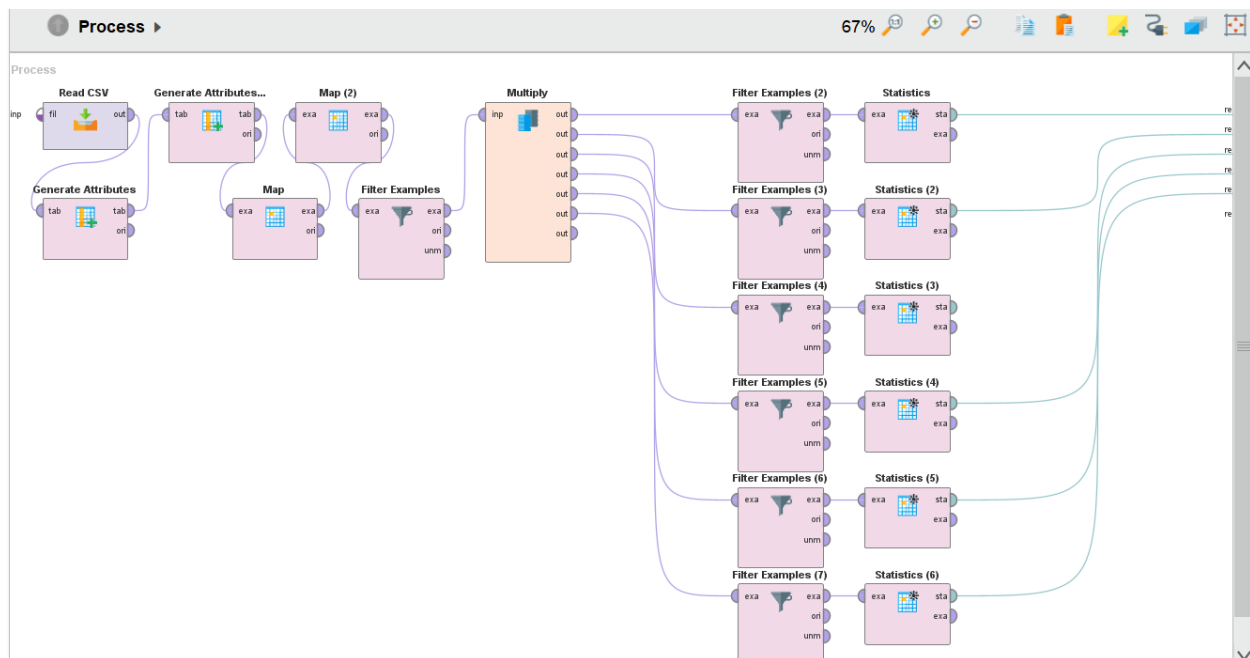


Figure 12: Process Diagram of Q3

Metric	Under-weight	Normal	Over-weight	Obesity I	Obesity II	Obesity III

Count	122	191	462	294	338	232
% Male	19.7%	50.3%	60.0%	59.9%	82.5%	0%
Avg Age	21.8	24.0	26.8	27.8	28.0	23.7
Avg ACTIVITY	1.18	1.22	0.99	1.00	0.82	0.64
Avg WATER	1.93	1.91	2.05	2.22	1.81	2.38
Avg UoT	0.68	0.59	0.59	0.61	0.56	0.57
Avg EVM	2.66	2.36	2.25	2.19	2.49	3.00
Avg BMI	17.6	22.3	27.2	32.4	37.2	42.8

Figure 13: Six Profiles Comparison Table

Trend	Observation
Physical Activity	Decreases as obesity increases (1.18 → 0.64)
Male %	Increases from 20% to 83% (except Obesity III – all female in this sample)
Vegetable Consumption	Highest in Underweight (2.66) and Obesity III (3.00)
Water Intake	Highest in Obesity III (2.38)

Figure 14: Key Insights from Profiles

Conclusion for Q3

As obesity level increases, physical activity decreases and male proportion increases. Obesity III shows unique patterns with highest vegetable and water intake.

Question 4: Lifestyle Clusters (K-Means)

Business Understanding

Goal: Find natural groupings of people based on lifestyle and health habits, independent of BMI, to identify segments for targeted interventions.

Data Understanding

Attributes used for clustering (BMI excluded): EVM, NDMM, EFBM, WATER, ACTIVITY, UoT, FIZZY, MONITOR, SMOKE, TRANSPORT.

Data Preparation

Z-Score normalization applied to all attributes. Adult-only filter applied (age ≥ 20).

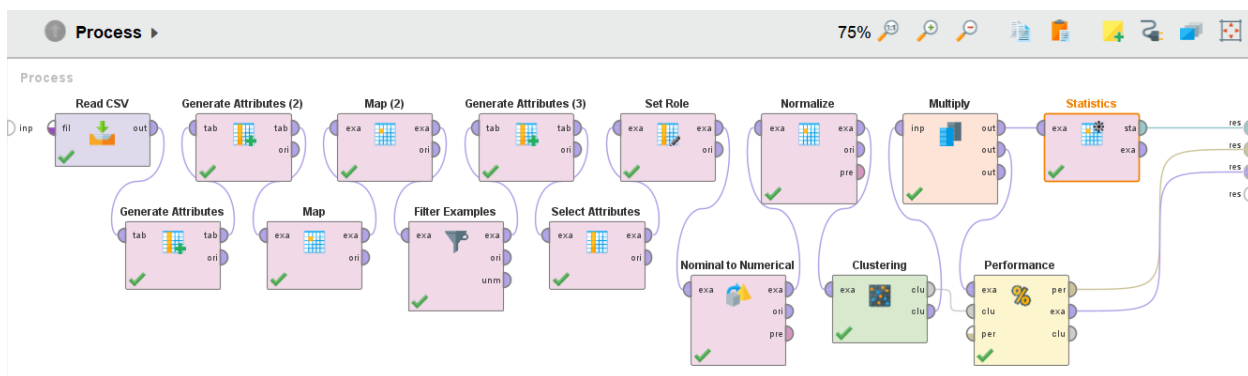


Figure 15: Process Diagram of Q4

Modeling

K-Means clustering. Optimal number of clusters determined by cluster size and Davies Bouldin index.

Optimal Number of Clusters

k	Cluster Sizes	Davies Bouldin	Decision
3	424, 33, 1,182	-1.571	Selected
4	267, 915, 447, 10	-1.924	Rejected (tiny 10-item cluster)
5	311, 242, 631, 422, 33	-1.662	Rejected (unbalanced)

Performance metrics for k=3:

- Avg. within centroid distance: -11.897

- Davies Bouldin Index: -1.571

Cluster Profiles (k=3)

Attribute	cluster_0	cluster_1	cluster_2
TRANSPORT = Public_Transportation	-1.608	-1.608	0.622
TRANSPORT = Walking	-0.143	6.974	-0.143
TRANSPORT = Automobile	1.655	-0.577	-0.577
TRANSPORT = Motorbike	0.224	-0.078	-0.078
TRANSPORT = Bike	0.142	-0.049	-0.049
EVM	-0.197	-0.024	0.071
NDMM	0.026	0.228	-0.016
WATER	-0.142	0.057	0.049
ACTIVITY	-0.003	0.642	-0.017
UoT	-0.232	0.159	0.079
EFBM_num	0	0	0
FIZZY_num	0	0	0
SMOKE_num	0.006	0.228	-0.008
MONITOR_num	0.009	0.584	-0.020

Figure 16: Centroid Table (Z-Score normalized values)

Result History	ExampleSet (Multiply)	PerformanceVector (Performance)	StatisticsIOObject (Statistics)
Performance Description Annotations	PerformanceVector PerformanceVector: Avg. within centroid distance: -11.897 Avg. within centroid distance_cluster_0: -17.224 Avg. within centroid distance_cluster_1: -17.350 Avg. within centroid distance_cluster_2: -9.834 Davies Bouldin: -1.571		

Figure 17: Performance vector

Summary Table – Three Lifestyle Clusters

Cluster	Name	Size	%	Transport	Activity	Diet	Health Status	Recommendation
0	Car-Dependent Moderate	424	25.9%	Automobile	Average	Below avg veg, water	At Risk	Promote active transport (walking/biking),

								increase vegetable and water intake
1	Active Healthy Walkers	33	2.0%	Walking	High	Low junk food, monitors calories	Healthiest	Preserve and study this group identify what enables their healthy habits
2	Public Transport Average	1,182	72.1%	Public Transport	Average	Average	Baseline	Targeted interventions to increase physical activity and reduce junk food

Key Insights from Clustering

Insight	Explanation
Only 2% are truly healthy	Cluster 1 represents just 33 out of 1,639 adults
Walking is a differentiator	The healthiest cluster defined by extremely high walking scores
Majority are average	72% fall into baseline cluster with room for improvement
Car dependence = risk	Cluster 0 shows below-average healthy behaviors
Healthy behaviors cluster together	High activity + low junk food + calorie monitoring + walking appear together

Conclusion for Question 4

K-Means clustering with **k=3** successfully identified three distinct lifestyle groups. The majority of the population (72%) falls into an "average" category with significant room for improvement. Only **2%** exhibit a truly healthy lifestyle characterized by walking, high physical activity, calorie monitoring, and low junk food consumption. Car dependence is associated with less healthy behaviors. Public health interventions should focus on promoting active transportation and reducing car dependency.

Conclusion

Summary of Findings

Question	Key Finding
Q1: BMI Factors	Eating between meals, overweight history, and transport choices most influence BMI
Q2: Obesity Factors	EFBM is the strongest predictor; HoV, gender, and calorie monitoring also critical
Q3: Six Profiles	Activity decreases, male % increases as obesity rises
Q4: Lifestyle Clusters	Only 2% exhibit truly healthy lifestyle patterns

Recommendations

Recommendation	Target	Expected Impact
Reduce eating between meals	All individuals	Lower BMI
Increase physical activity	Clusters 0 and 2	Move to healthier clusters
Reduce fizzy drink consumption	All individuals	Lower obesity risk
Promote calorie monitoring	Clusters 0 and 2	Better weight management
Encourage walking/biking	Automobile users	Lower BMI, healthier lifestyle

Limitations

Limitation	Impact
Self-reported data	Potential bias
Young adult skew	May not generalize to older populations
Small sample for some BMI categories	Obesity I had only 294 cases

No longitudinal data	Cannot infer causation
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Lessons Learned

1. Data preparation is critical, Ordinal variables must be properly converted
2. Class imbalance affects model performance, Obesity I had only 2% recall
3. Clustering without BMI reveals true lifestyle patterns, only 2% are truly healthy
4. CRISP-DM framework ensures structured analysis
5. Altair AI Studio effectively handles end-to-end data mining

References

1. Altair AI Studio Documentation (2026) <https://altair.com/altair-ai-studio>
2. CRISP-DM Framework Guide
https://www.ibm.com/docs/it/SS3RA7_18.3.0/pdf/ModelerCRISPDM.pdf
3. WHO Obesity Classification Standards <https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight>